FCFS Appointment Simulation:
Metric-Focused Sensitivity Analysis

Supervisor Review Version

CUIMC Appointment Simulation

May 9, 2026

Abstract

A first-come, first-served (FCFS) outpatient appointment simulation with two symmetric patient classes was run under a wide range of behavioral assumptions. The central finding is that utilization and access are distinct outcomes: at baseline, the system uses 83.9% of available slots but serves only 26.9% of arriving patients. Total demand pressure is the strongest driver of overall served rate, while class-specific behavioral differences—primarily cancellation probability, balking threshold, and no-show step—are the dominant drivers of class disparity. This memo explains the metric hierarchy, reports the main sensitivity findings, and identifies the model's key limitations.

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1 Research Question and Interpretation Framework

Research Question

How does FCFS appointment scheduling perform across a range of demand and behavioral assumptions, and which patient-level behavioral parameters most strongly affect access and equity?

Why Multiple Metrics Are Needed

A single metric cannot characterize appointment system performance because utilization, patient access, patient-facing wait, and class equity are logically independent outcomes. A clinic can be fully booked (high utilization), yet serve a small fraction of requesting patients (low access), if many patients balk at long offered waits or cancel after booking. Similarly, average wait times can appear short if patients with long offered delays are systematically excluded from the accepted-booking pool.

The simulation therefore tracks five conceptually distinct outcomes:

1. **Capacity use**: how busy is the clinic?
2. **Patient access**: what share of patients who want service actually receive it?
3. **Patient-facing wait**: how long must an average arriving patient wait?
4. **Rejection behavior**: how often do patients reject offered appointments, and why?
5. **Class equity**: do different patient groups experience systematically different access?

Each metric is necessary to interpret the others, and none is sufficient on its own.

Metric Hierarchy

Metric	What it measures	How to interpret it	Main caveat
Overall served rate	Share of arriving patients who complete service	Primary access metric; lower is worse	Denominator is all arrivals, not just those who book
Average utilization	Share of clinic slots filled with completed visits	Capacity-use metric; high does not imply good access	No-shows and balking inflate utilization relative to access
Mean offered wait	Average delay offered to all patients who receive a slot offer	Patient-facing wait, inclusive of those who then balk	Harder to interpret when demand mix changes
Mean accepted wait	Average delay among patients who accepted a booking	Conditional delay metric	Selection bias: excludes patients who reject long offers
Balking rate	Share of offered patients who reject the offered slot	Congestion and rejection diagnostic	Not a success metric; lower is not automatically better
Class served-rate gap	Class 1 served rate minus Class 2 served rate	Fairness / class disparity metric	Meaningful only when class behavioral parameters differ

Table 1: Metric hierarchy. Primary access metric is overall served rate; all others contextualize it.

2 Baseline Diagnosis

2.1 Simulation Setup

The baseline uses `configs/baseline.yaml`: 32 slots per day, a 14-day booking horizon, two symmetric classes each with 50 arrivals per day, cancellation probability 0.10, balking threshold 9 days with high-delay balking probability 0.50, and no-show threshold 6 days with high-delay no-show probability 0.30. Low-delay balking and no-show probabilities are both 0.00. The measurement window is 365 days following a 30-day burn-in.

2.2 Baseline Metric Summary

Avg. utilization	Overall served	Mean accepted wait	Mean offered wait	Class gap	Delay gap
0.839	0.269	8.35	9.30	0.001	0.003

Table 2: Baseline metric outcomes. Class gap is Class 1 served rate minus Class 2 served rate. Delay gap is Class 2 mean offered wait minus Class 1 mean offered wait.

The key diagnostic is the utilization–access gap. At baseline, the system uses 83.9% of available slots but serves only 26.9% of arriving patients. This gap indicates that the system is effectively capacity-constrained: the booking horizon fills quickly, large numbers of patients receive no slot offer or balk at long offered delays, and after-booking losses (cancellations and no-shows) further reduce completed visits. A facility-manager who only monitors utilization would conclude the clinic is functioning well; the patient-side metrics tell a different story.

3 Sensitivity Analysis

3.1 Overview

The sensitivity analysis varies one group of assumptions at a time around the baseline to isolate how each driver moves each metric. All runs use the same FCFS engine; only the input parameters and random seeds change. Each result is averaged across multiple seeds to reduce sampling noise. The analysis is a model-based stress test, not an empirical causal estimate.

3.2 Average Utilization

Definition

$$\texttt{average\_utilization} = \frac{1}{D}\sum_{d}\frac{\text{served slots on day } d}{\text{slots per day}}.$$

No-shows do not count as utilized slots because the visit was not completed.

Main Drivers and Interpretation

Figure 1 shows that no-show behavior is the clearest direct driver of utilization: booked slots that become no-shows never convert to completed visits. Cancellation also matters because it changes the appointment pool before the service day. Demand pressure, by contrast, has a weaker effect on utilization once the system is already operating near capacity—additional arrivals are simply turned away or balk, leaving the filled slots largely unchanged.

The operational implication is that a provider concerned primarily about utilization should focus on no-show reduction and cancellation management rather than on reducing demand. However, high utilization achieved by filling slots with patients who would have been lost to no-shows or cancellations does not directly translate into better patient access.

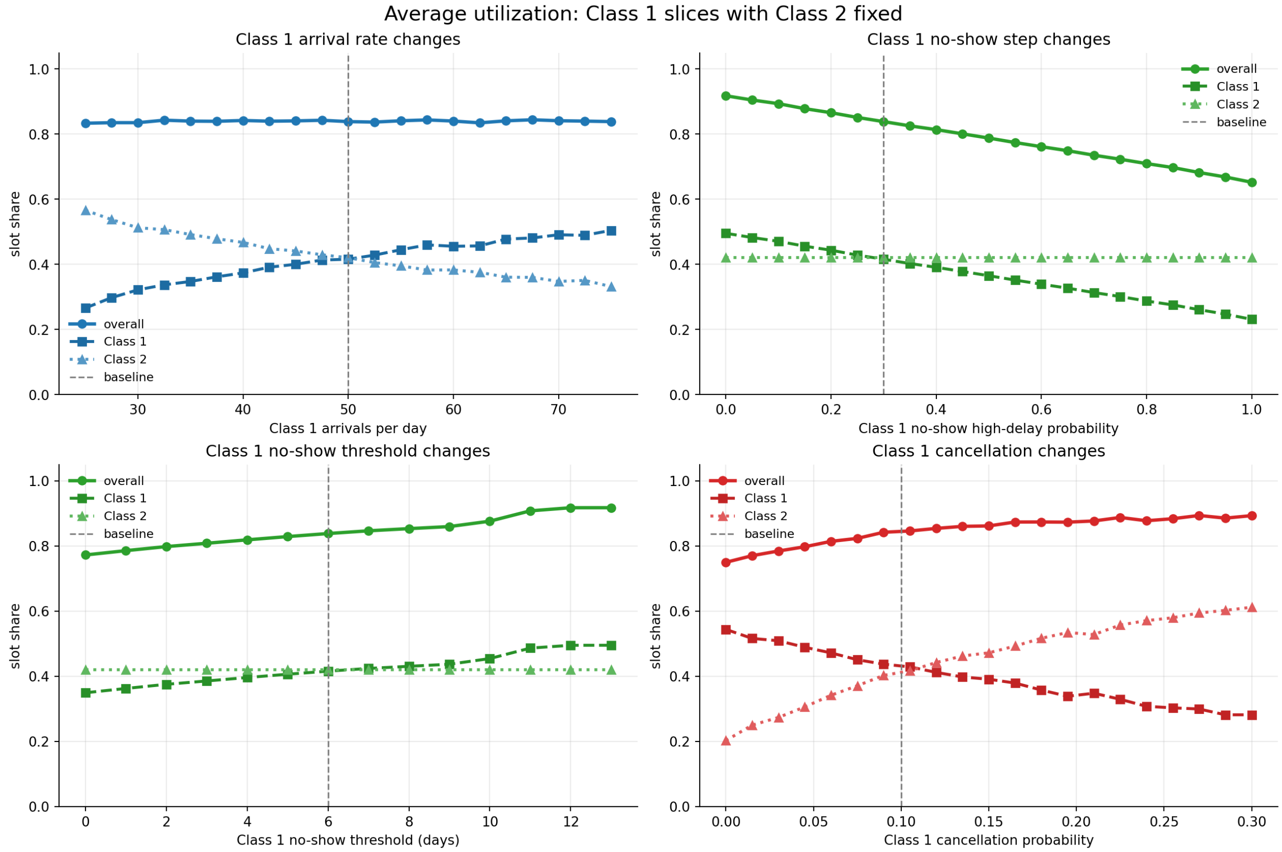

Figure 1: Utilization remains near 0.839 across a wide demand range, while overall served rate falls sharply—confirming that utilization is a poor proxy for patient access. Class 1 varies on the x-axis; Class 2 stays at baseline.

3.3 Overall Served Rate

Definition

$$\texttt{overall\_percent\_serviced} = \frac{\sum_i V_i}{\sum_i A_i},$$

where A_i is arrivals and V_i is completed visits for class i. This is the primary access metric.

Main Drivers and Interpretation

Figure 2 shows that total arrival pressure is the dominant aggregate driver. As demand rises against fixed 32-slot daily capacity and a 14-day booking horizon, an increasing share of arrivals either receives no slot offer (the horizon is full) or balks at long offered delays. No-shows and cancellations reduce completed visits after booking.

The regression screen (Section 5) confirms this ordering: total arrival rate has the largest standardized coefficient for overall served rate ($\beta = -0.774$), followed by average no-show threshold ($\beta = 0.251$) and no-show step ($\beta = -0.175$).

Near the baseline, the system is already capacity-constrained: virtually all slots are occupied yet fewer than 30% of patients are served. This means that incremental increases in demand primarily worsen unmet access rather than further increasing utilization. Conversely, modest reductions in arrival load—or increases in daily capacity—would have an outsized effect on served rate in this regime.

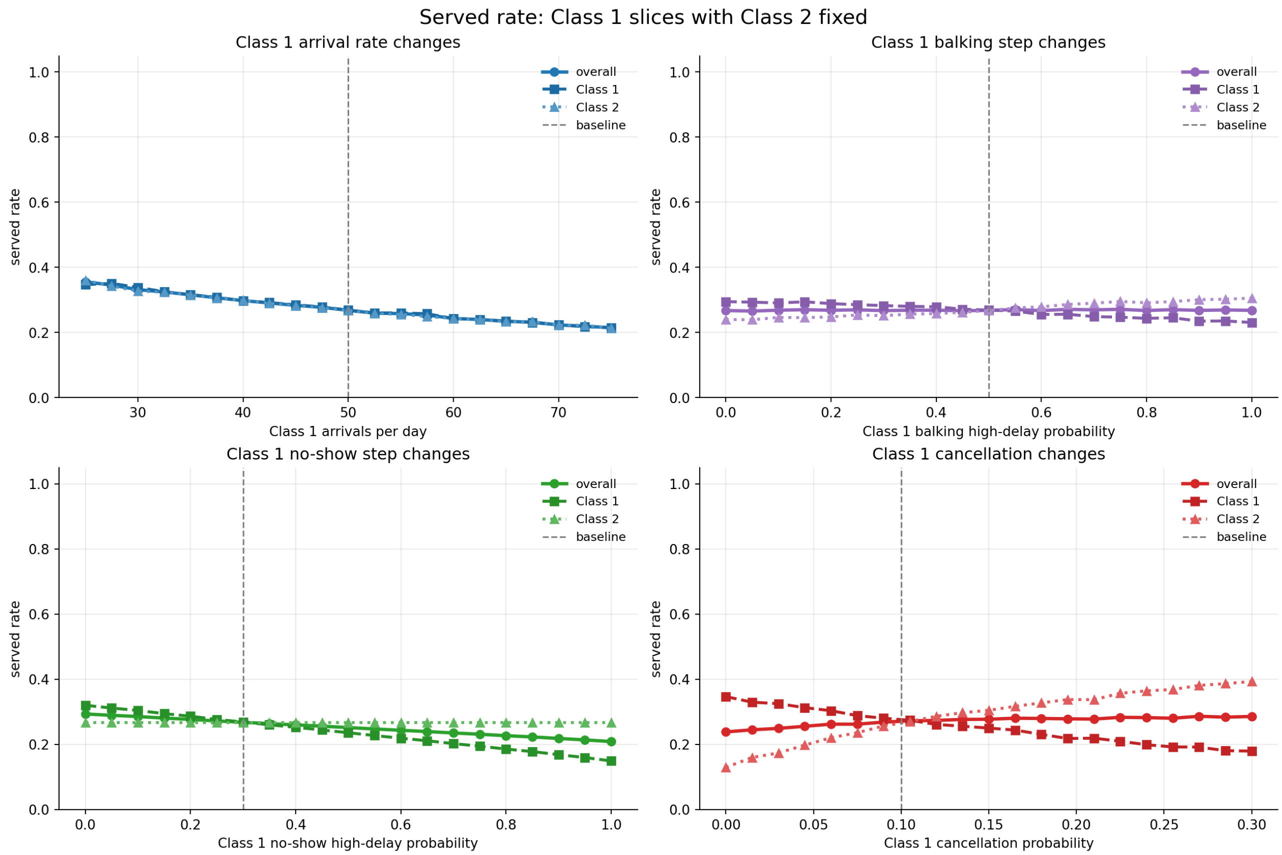

Figure 2: Overall served rate falls as total demand or class-specific arrival rates rise, even when utilization stays high. At baseline, 50 arrivals per class per day produces a served rate near 0.269, well below the utilization of 0.839.

3.4 Mean Offered Wait

Definition

$$\texttt{mean\_offered\_booking\_delay} = \frac{\sum_i \text{total offered delay}_i}{\sum_i O_i},$$

where $O_i = B_i + K_i$ is the number of offered patients: those who accepted plus those who balked. Patients who received no slot offer are excluded because no delay was presented to them.

Why Offered Wait Is Preferred Over Accepted Wait

Mean accepted wait is computed only over patients who accepted a booking. When balking rates are high, the accepted sample is systematically selected against patients who received—and rejected—long-delay offers. Accepted wait therefore understates the delay that arriving patients face.

Mean offered wait corrects for this selection by including balked patients. It answers the question: among patients who received any delay offer, what was the average delay presented? Figure 3 shows that offered wait is more sensitive to demand pressure than accepted wait, because rising demand pushes more offers to the far end of the booking horizon before patients reject them.

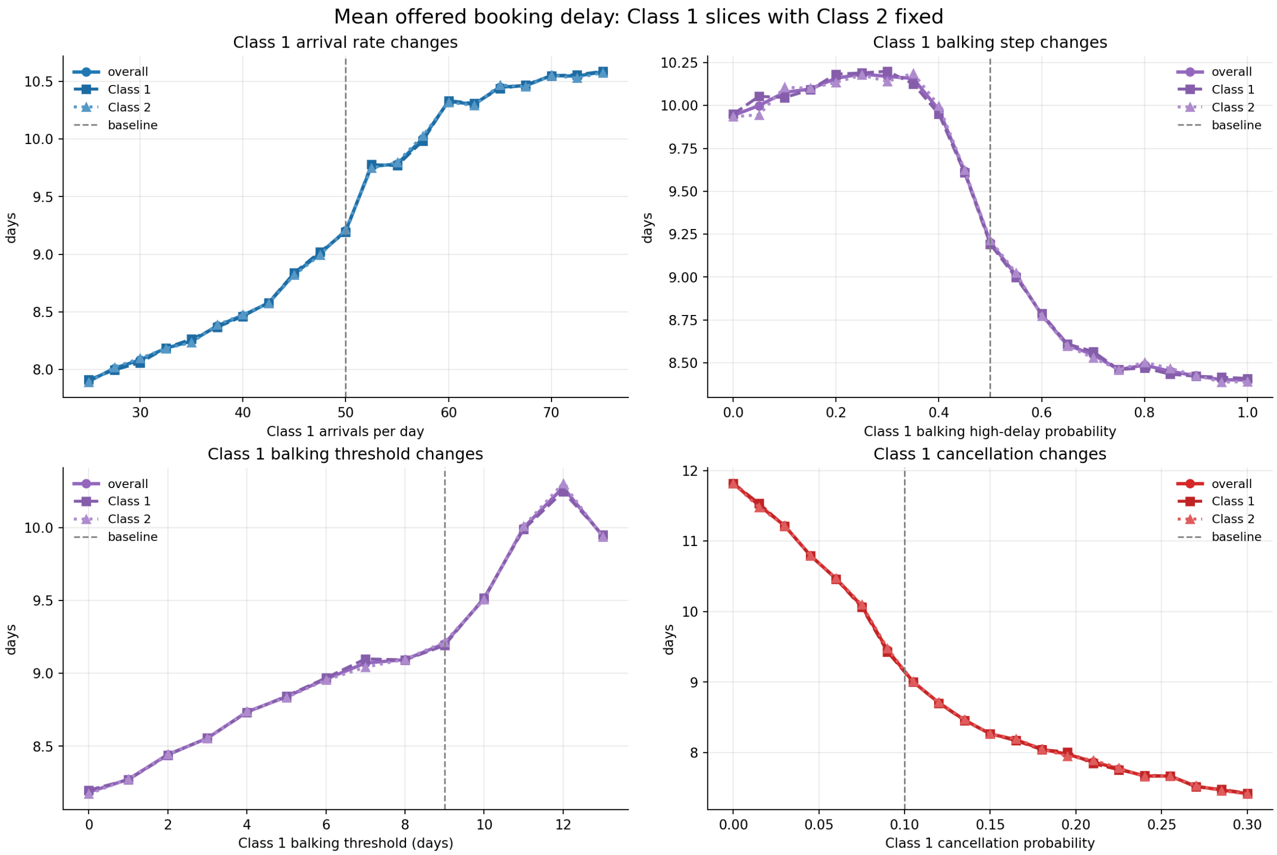

Figure 3: Mean offered wait rises with demand pressure and falls when balking selects out long-delay patients. Mean accepted wait can fall even as access worsens because patients who reject long-delay offers are removed from the accepted sample.

The regression screen finds total arrival rate ($\beta = 0.576$) as the dominant driver of offered wait, followed by average cancellation probability ($\beta = -0.441$, note: cancellations free future capacity and reduce congestion), average balking threshold ($\beta = 0.278$), and average balking step ($\beta = -0.206$). The negative sign on balking step is a selection effect: higher step means more long-delay patients reject offers, which removes them from the offered-wait average.

3.5 Balking Rate

Definition

$$\texttt{balking\_rate} = \frac{\sum_i K_i}{\sum_i O_i}.$$

Measured among patients who received an offer. Patients who received no offer are excluded.

Interpretation as a Diagnostic

Balking rate measures how often patients reject offered appointments. It is a useful congestion diagnostic—rising balking rate indicates that offered delays are routinely exceeding patient tolerance—but it is not a success metric. A lower balking rate does not imply better access if it results from fewer patients receiving offers, reduced arrival rates, or a selection shift in who presents.

Figure 4 shows that balking rate is controlled mainly by the balking step (probability increase above threshold) and the threshold itself. The key interpretive warning is that lower accepted wait often accompanies higher balking: when more long-delay patients leave the system, the remaining accepted bookings are shorter on average. This is a selection effect, not a service improvement.

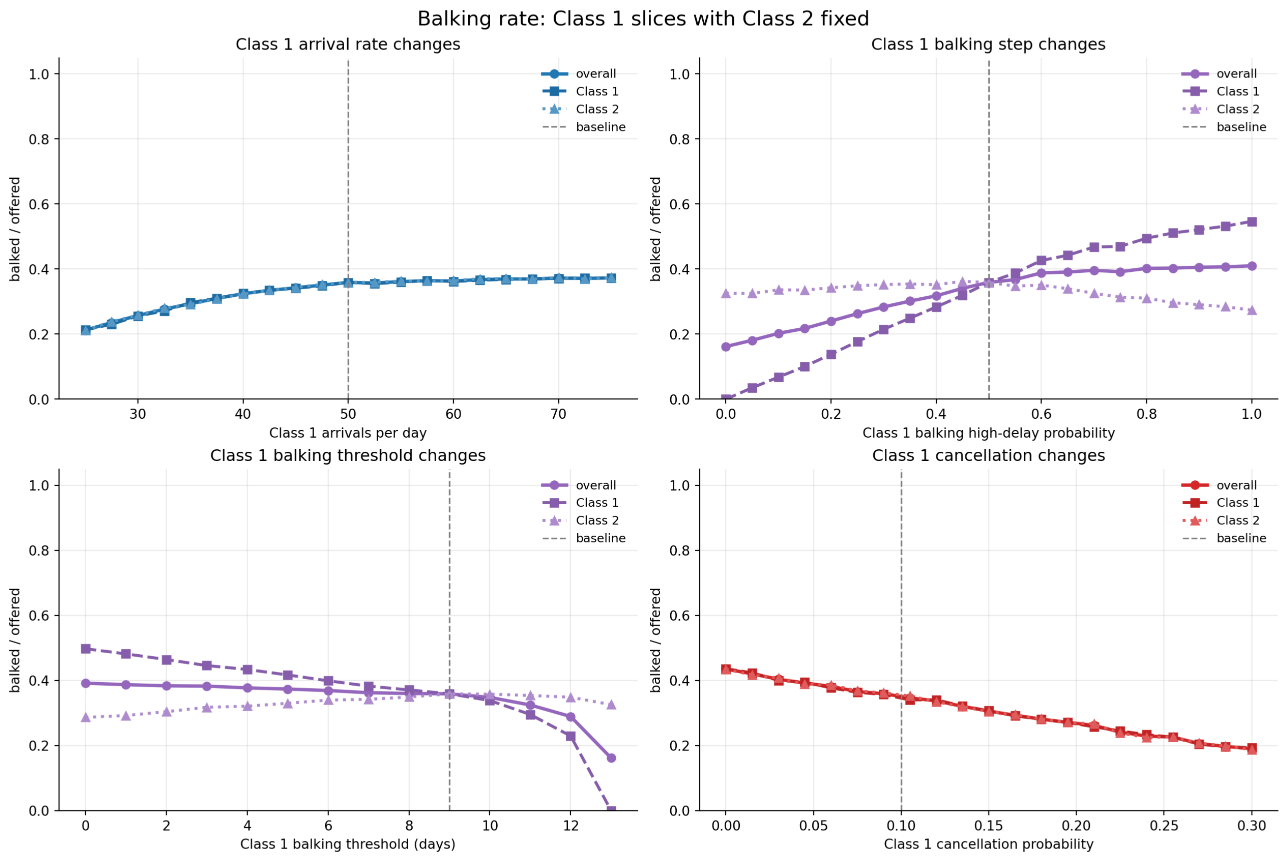

Figure 4: Balking rate rises with the high-balking probability and falls with the threshold. Higher balking can coexist with lower accepted wait, because rejecting long-delay patients removes them from the accepted sample without reducing offered delays for remaining patients.

3.6 Class Served-Rate Gap

Definition

$$\Delta_{\text{access}} = \texttt{percent\_serviced}_1 - \texttt{percent\_serviced}_2.$$

Positive values indicate Class 1 is served more often than Class 2.

When Class Labels Matter

In the FCFS baseline, both classes have identical behavioral parameters. Class labels are therefore exchangeable: the expected class gap is zero, and observed deviations are noise. Class disparity becomes meaningful only when the classes receive different behavioral assumptions.

Figure 5 illustrates this directly: holding Class 2 fixed at baseline while varying Class 1 assumptions, the class gap widens as Class 1 behavioral parameters diverge from Class 2. Higher Class 1 cancellation probability, higher balking step, or higher no-show step all move the gap against Class 1.

The regression screen (Section 5) ranks the gap drivers: cancellation probability gap ($\beta = -0.452$), balking threshold gap ($\beta = 0.429$), balking step gap ($\beta = -0.349$), no-show threshold gap ($\beta = 0.268$), and no-show step gap ($\beta = -0.211$) are all significant. Class 1 arrival share ($\beta = -0.120$) also enters significantly, reflecting that FCFS is sensitive to the temporal ordering of arrivals as well as their count.

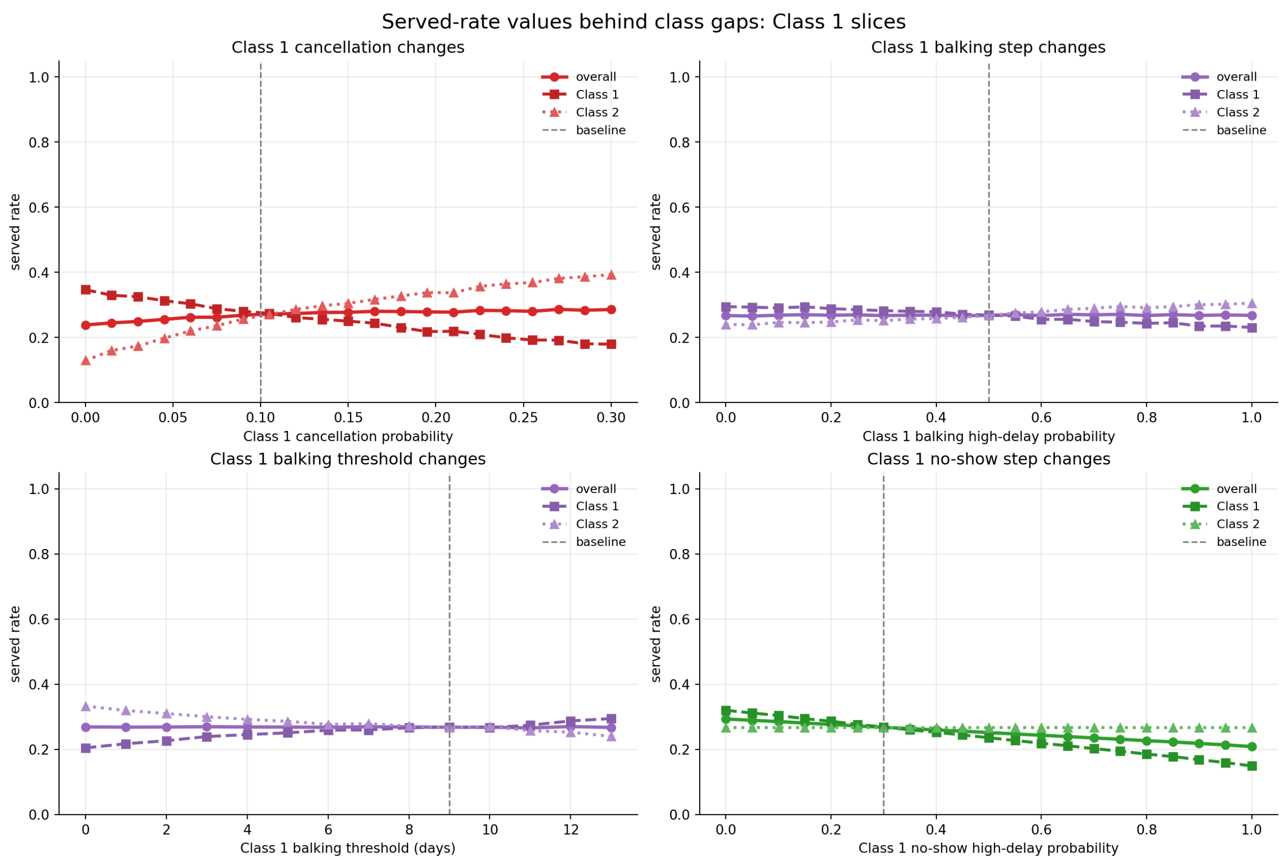

Figure 5: The class served-rate gap is driven by differences in behavioral parameters. With symmetric assumptions, both class lines track together. As Class 1 behavioral parameters diverge from Class 2 (e.g., higher cancellation or balking), Class 1 access falls below Class 2.

4 Balking Deep Dive

This section isolates Class 1 balking behavior while Class 2 stays at baseline. The purpose is to demonstrate the selection mechanism that makes lower accepted wait a potentially misleading signal.

4.1 Class 1 High Balking Probability

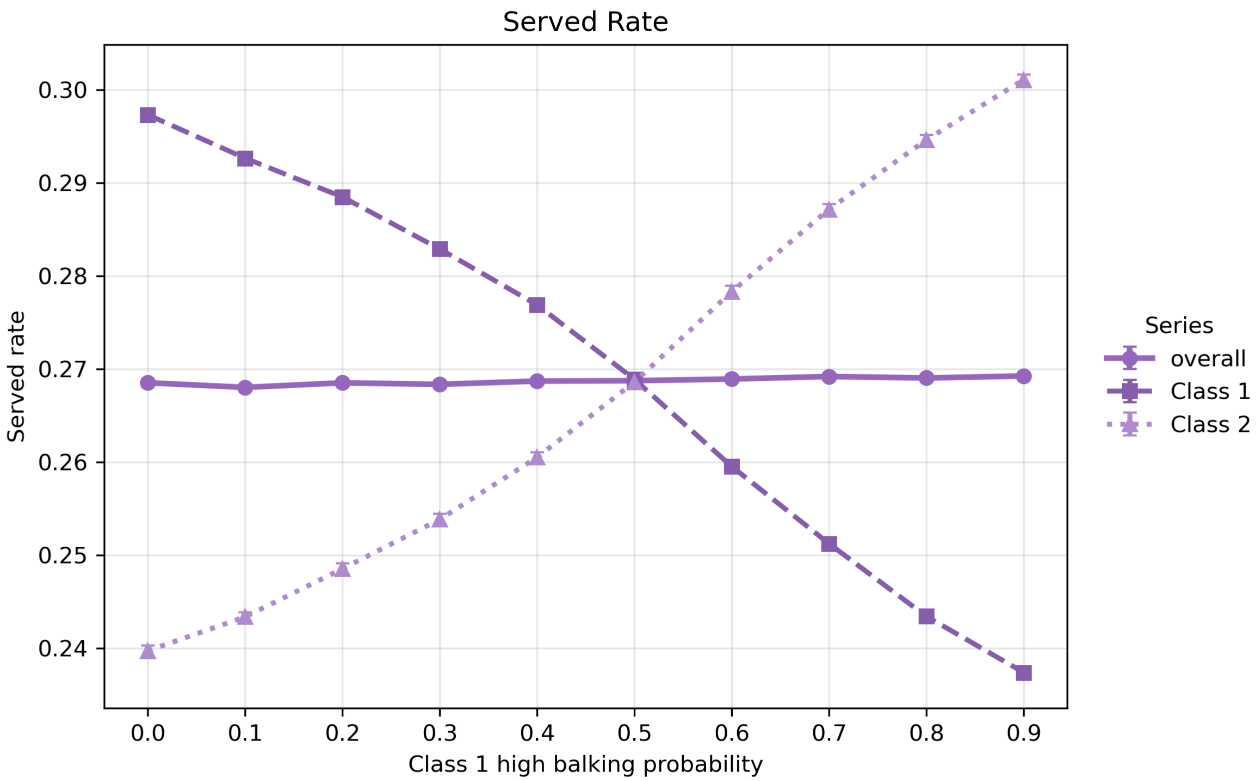

Figure 6: As Class 1 high-balking probability increases, Class 1 access falls while Class 2 access rises. The redistribution shows that higher balking worsens access for the more-balking class and partially relieves congestion for the other.

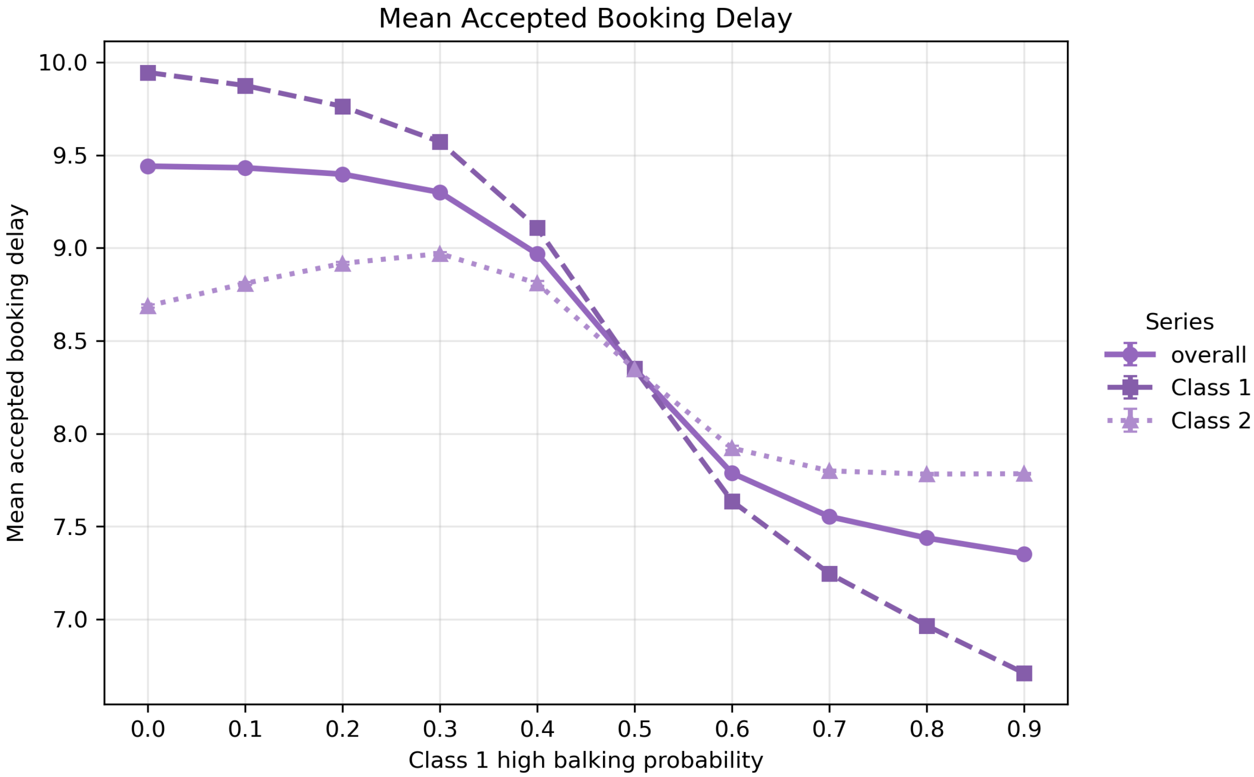

Figure 7: Class 1 accepted delay falls as high-balking probability increases, but this is a selection effect: long-delay Class 1 patients leave the accepted sample, not because slots become available sooner.

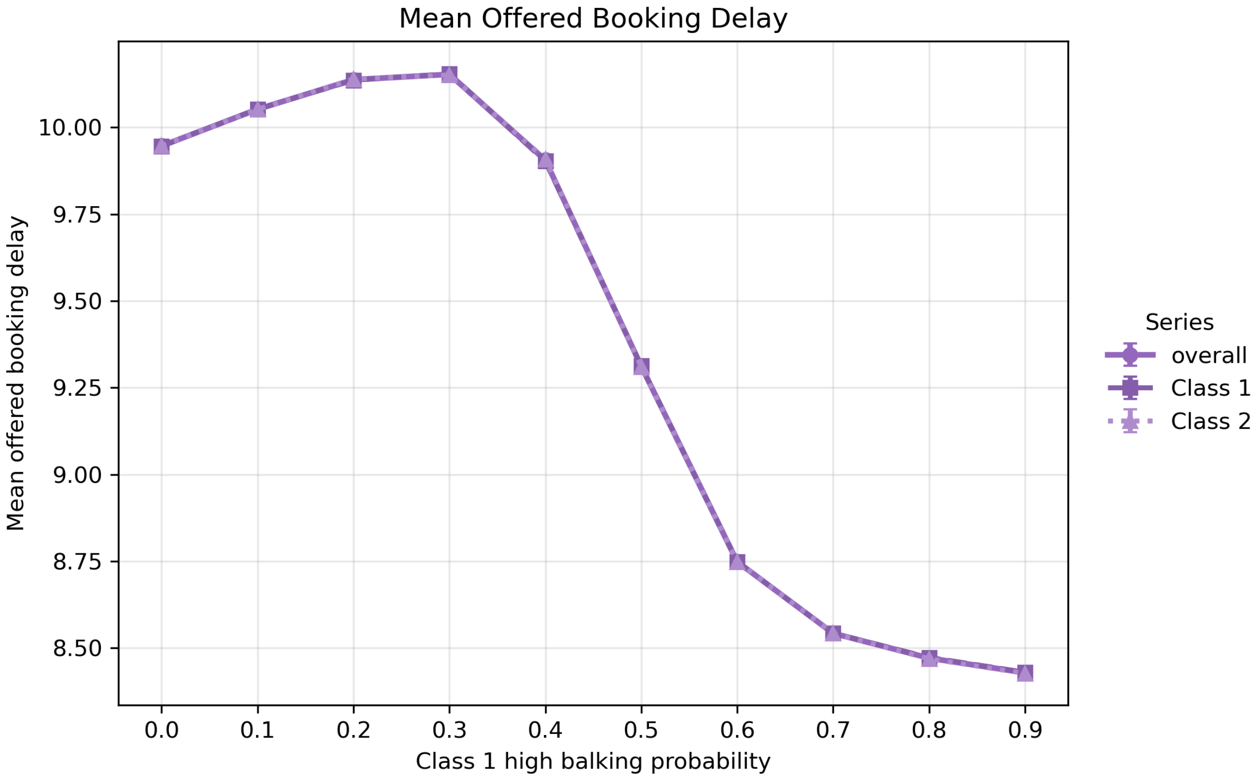

Figure 8: Mean offered delay captures the broader congestion effect. Once enough long-delay Class 1 patients reject offers, congestion falls and offered delays shorten for both classes.

The three figures together demonstrate that accepted wait and served rate can move in opposite directions when balking changes. A policy intervention that increases balking step would appear to improve Class 1 wait times while actually worsening Class 1 access.

4.2 Class 1 Balking Threshold

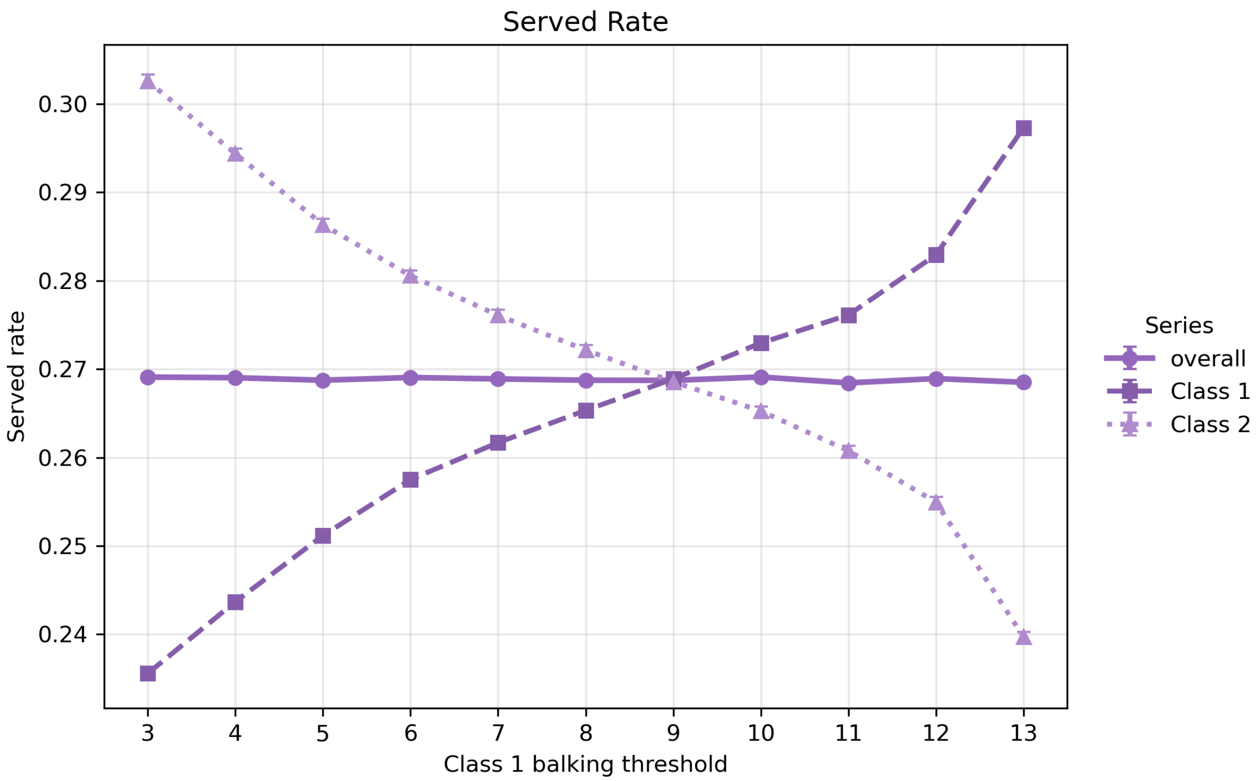

Figure 9: Raising Class 1's balking threshold increases Class 1 access by tolerating longer offered waits. The benefit comes at the cost of slightly longer congestion for Class 2 when the threshold is very high.

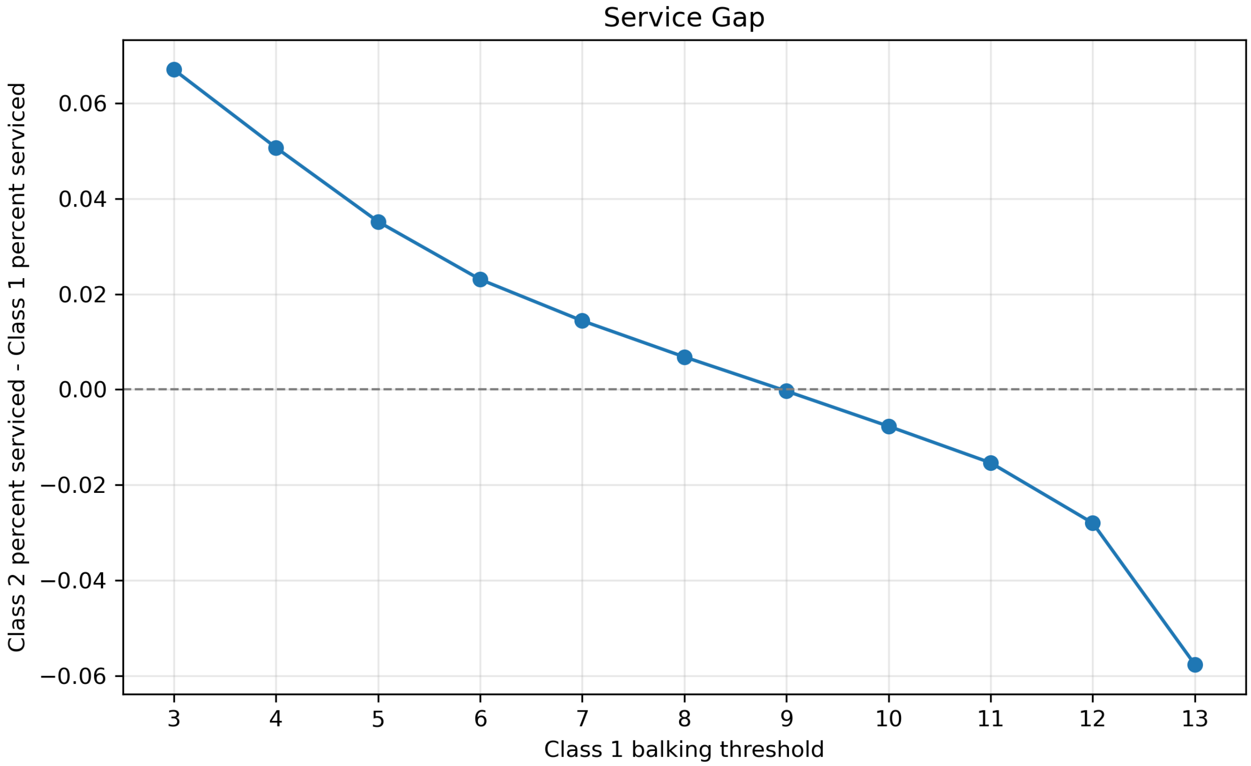

Figure 10: The Class 1 minus Class 2 service gap is positive when Class 1 has a higher threshold, reflecting that the more tolerant class captures more available slots in FCFS.

5 Regression Screen

The regression screen tests whether the main visual findings hold when all assumptions vary simultaneously. For each of 240 randomized FCFS parameter settings, the simulation is run with two seeds and the outputs averaged. Twelve features are constructed from the sampled inputs: total arrival rate, Class 1 arrival share, average levels, and Class 1 minus Class 2 gaps for balking step, balking threshold, no-show step, no-show threshold, and cancellation probability. Four targets are modeled separately.

All variables are standardized before fitting ordinary least squares. Significance is assessed using HC3 robust standard errors; only coefficients with robust p-values below 0.05 are reported. Full-sample R^2 ranges from 0.639 (class access advantage) to 0.799 (overall served rate), and out-of-sample R^2 from 0.545 (utilization) to 0.719 (overall served rate), indicating that the linear screen explains meaningful variation in the simulation design.

These are simulation associations within the randomized design, not causal estimates from real appointment data.

Target metric	Significant feature	Std. coef.	95% HC3 CI	HC3 p
Utilization	Average no-show threshold	+0.512	[0.433, 0.591]	< 0.001
Utilization	Average no-show step	−0.423	[−0.501, −0.346]	< 0.001
Utilization	Average cancellation probability	+0.320	[0.247, 0.393]	< 0.001
Utilization	Total arrival rate	−0.109	[−0.188, −0.031]	0.006
Utilization	Average balking threshold	−0.086	[−0.162, −0.009]	0.028
Overall served rate	Total arrival rate	−0.774	[−0.855, −0.694]	< 0.001
Overall served rate	Average no-show threshold	+0.251	[0.193, 0.308]	< 0.001
Overall served rate	Average no-show step	−0.175	[−0.232, −0.119]	< 0.001
Overall served rate	Average cancellation probability	+0.117	[0.058, 0.176]	< 0.001
Offered wait	Total arrival rate	+0.576	[0.499, 0.653]	< 0.001
Offered wait	Average cancellation probability	−0.441	[−0.506, −0.376]	< 0.001
Offered wait	Average balking threshold	+0.278	[0.202, 0.353]	< 0.001
Offered wait	Average balking step	−0.206	[−0.291, −0.122]	< 0.001
Class gap	Cancellation probability gap	−0.452	[−0.558, −0.347]	< 0.001
Class gap	Balking threshold gap	+0.429	[0.331, 0.528]	< 0.001
Class gap	Balking step gap	−0.349	[−0.431, −0.266]	< 0.001
Class gap	No-show threshold gap	+0.268	[0.179, 0.357]	< 0.001
Class gap	No-show step gap	−0.211	[−0.286, −0.136]	< 0.001
Class gap	Class 1 arrival share	−0.120	[−0.209, −0.030]	0.009

Table 3: Significant regression drivers ($p < 0.05$, HC3 robust). Gap features are Class 1 value minus Class 2 value. Coefficients are in standard deviation units.

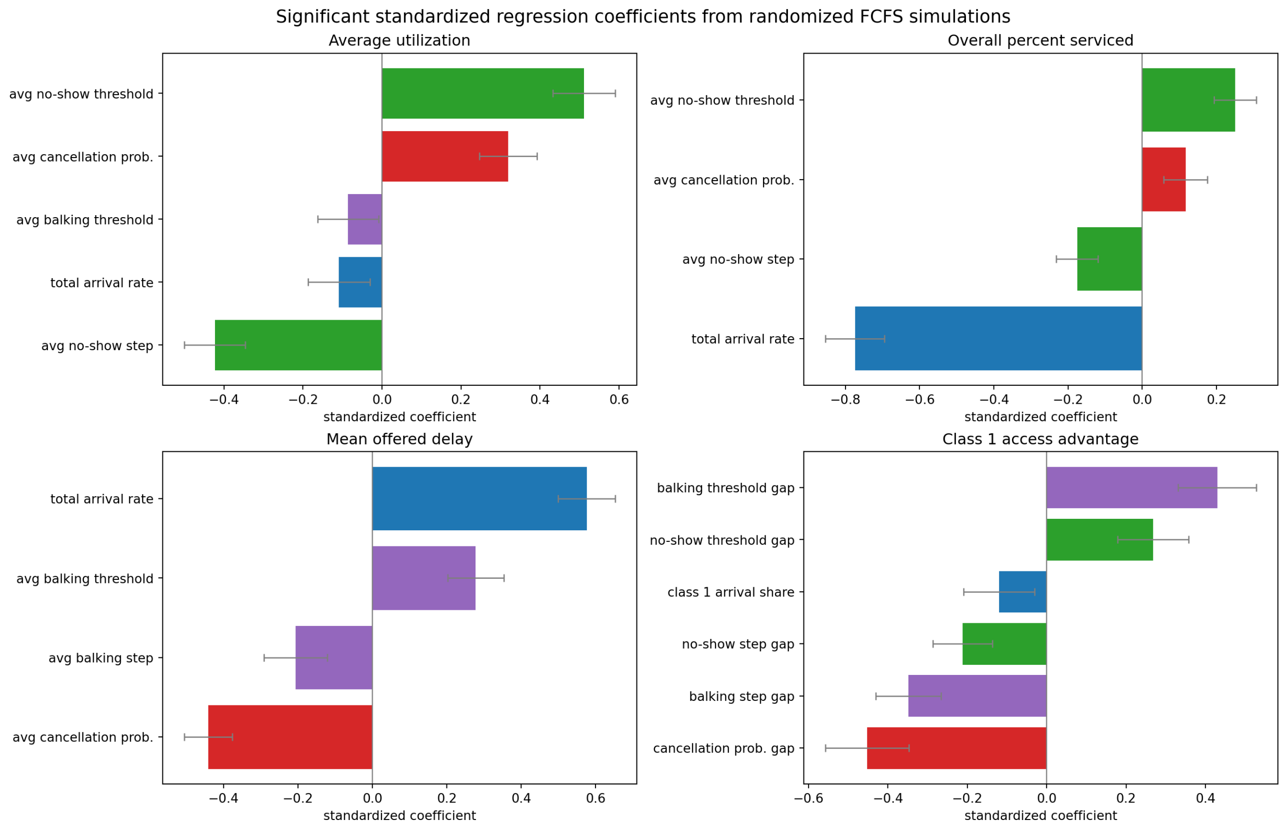

Figure 11: Regression coefficients confirm the metric-by-metric reading: no-show behavior dominates utilization, demand pressure dominates served rate and offered wait, and class behavioral gaps dominate the class served-rate gap. Error bars are 95% HC3 robust confidence intervals.

6 Key Findings

1. **High utilization can coexist with poor access.** At baseline, utilization is 0.839 but overall served rate is 0.269. The system fills available slots while failing to serve the majority of arriving patients. Utilization should not be used as a standalone performance metric in this scheduling context.

2. **Overall served rate is the cleanest access metric.** It uses all arrivals as the denominator and counts only completed visits. It is not affected by balking selection or the distinction between offered and accepted delays.

3. **Mean offered wait is preferred over mean accepted wait for patient-facing delay.** Accepted wait is conditional on patients accepting the offered appointment. It falls mechanically when patients with long offered delays balk, giving the misleading impression that congestion has improved. Offered wait includes those patients and better reflects the delay that the system actually imposes.

4. **Balking can mask congestion.** The deep-dive analysis shows that increasing Class 1 balking probability reduces Class 1 accepted wait while simultaneously reducing Class 1 access and potentially reducing overall offered wait. Interpreting lower accepted wait as a service improvement in this setting is incorrect.

5. **Demand pressure is the dominant driver of access; behavioral parameters dominate class equity.** The regression screen assigns the largest absolute standardized coefficient for overall served rate to total arrival rate ($|\beta| = 0.774$). For class served-rate gap, the largest coefficient belongs to the cancellation probability gap ($|\beta| = 0.452$), followed by balking threshold gap and balking step gap.

6. **Class disparities arise from behavioral asymmetries, not from class labels.** In the symmetric baseline, the class gap is near zero (0.001). Gap magnitude and direction track differences in cancellation probability, balking threshold, balking step, and no-show parameters. Equity-relevant interventions should therefore target behavioral parameter differences rather than class membership per se.

7. **Regression associations are simulation-internal and should not be interpreted as real-world causal effects.** The randomized screen shows stable linear associations within the simulation design. Real-world behavioral parameters are not observed; they are assumed. Translating these findings to clinic policy requires empirical calibration.

7 Limitations and Next Steps

7.1 Limitations

1. **Stylized simulation design.** The model uses threshold-type behavioral rules with sharp transition probabilities. Real patients likely have smoother, heterogeneous delay tolerances. The sharp thresholds amplify sensitivity around the threshold values in ways that may not appear in observed data.

2. **Assumed behavioral parameters.** Balking probabilities, no-show probabilities, and cancellation rates are set by assumption, not estimated from patient data. The sensitivity analysis covers a wide parameter range precisely because the true values are unknown. Any policy inference requires calibration against clinic records.

3. **Limited replication.** Most sensitivity grid points use one to three random seeds. Metrics with high seed-to-seed variance (particularly the class gap at small parameter differences) may not have converged. The regression screen averages only two seeds per setting.

4. **Single scheduling discipline.** The entire analysis uses FCFS. FCFS is a reasonable baseline but is not the only feasible policy. Priority-based, panel-based, or advanced-access scheduling may have qualitatively different utilization–access tradeoffs.

5. **No provider-side dynamics.** The model holds slots per day fixed. Real clinics adjust templates, add sessions, or use overbooking. These supply-side responses are not modeled.

7.2 Next Steps

1. **Calibrate behavioral parameters from clinic data.** Estimate patient-specific balking and no-show rates using appointment history and no-show records. Use these estimates to anchor the sensitivity grid around empirically plausible ranges.

2. **Evaluate alternative scheduling disciplines.** Implement and compare at least one priority-based policy (e.g., by clinical urgency or social risk) to assess whether FCFS produces systematically worse access for higher-need classes.

3. **Increase simulation replications.** Run at least 10–20 seeds per grid point for the primary metrics and at least 30–50 for the class gap, which is more variable. Report confidence intervals on metric estimates.
4. **Model supply-side feedback.** Introduce session-level cancellation and overbooking rules to test how the clinic-side response to no-shows affects the utilization–access gap.
5. **Extend to provider panel and network effects.** If patient panels are partitioned across providers, FCFS within a panel may produce different access outcomes than pooled FCFS. Modeling panel structure would allow equity analysis at the provider level.

A Appendix: Supplementary Sensitivity Plots

These figures are retained for traceability. They support the main text but are not required for the primary findings.

A.1 No-Show Sensitivity

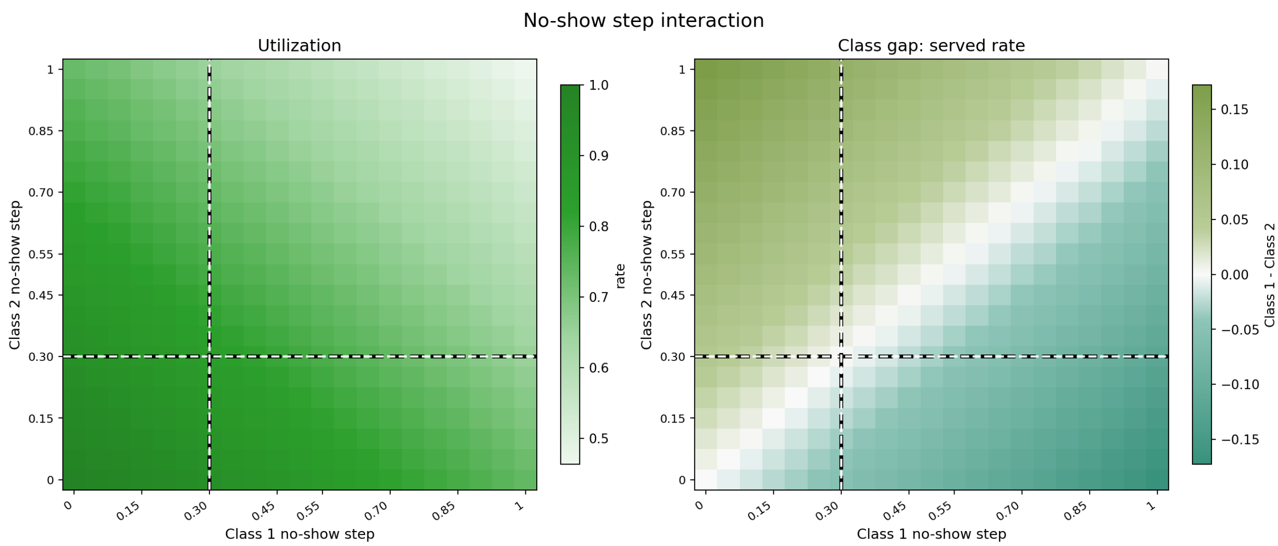

Figure 12: No-show step heatmap. Left: utilization falls when both classes have high no-show risk. Right: class gap moves against the class with higher no-show step.

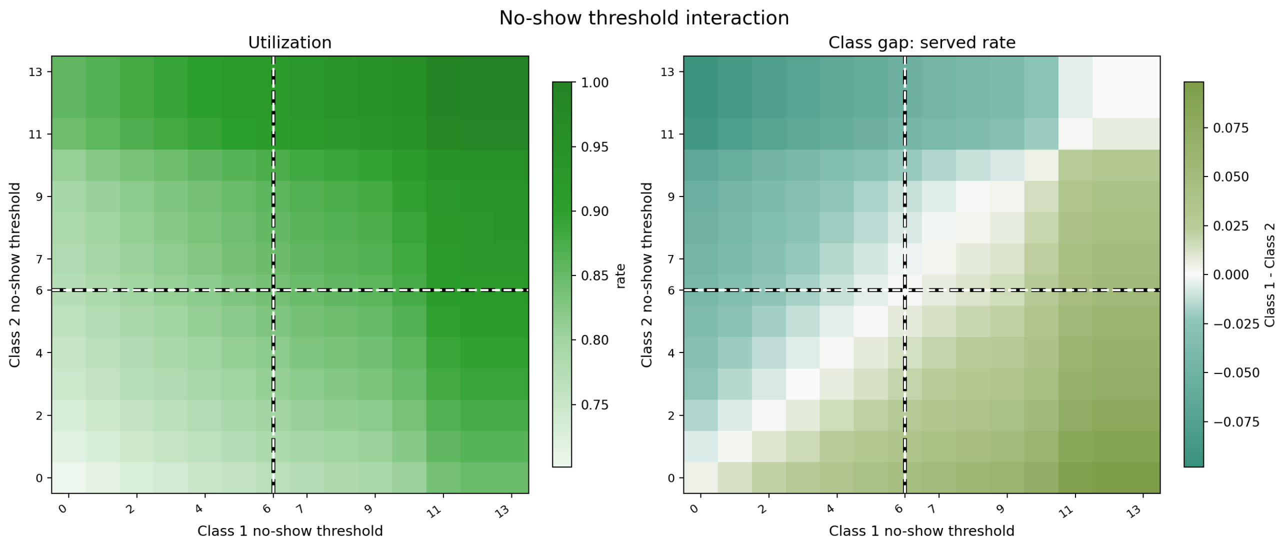

Figure 13: No-show threshold heatmap. A higher threshold delays the onset of high no-show risk; utilization and access improve for the affected class.

A.2 Balking Sensitivity

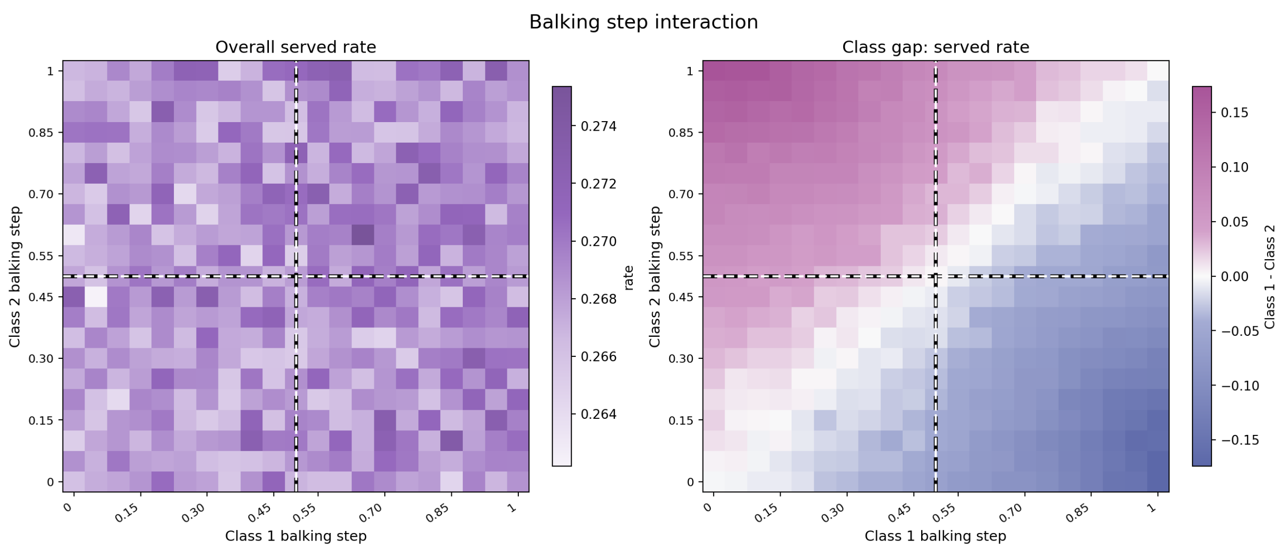

Figure 14: Balking step heatmap. Left: overall served rate falls as either class increases its high-balking probability. Right: class gap moves against the class with higher balking step.

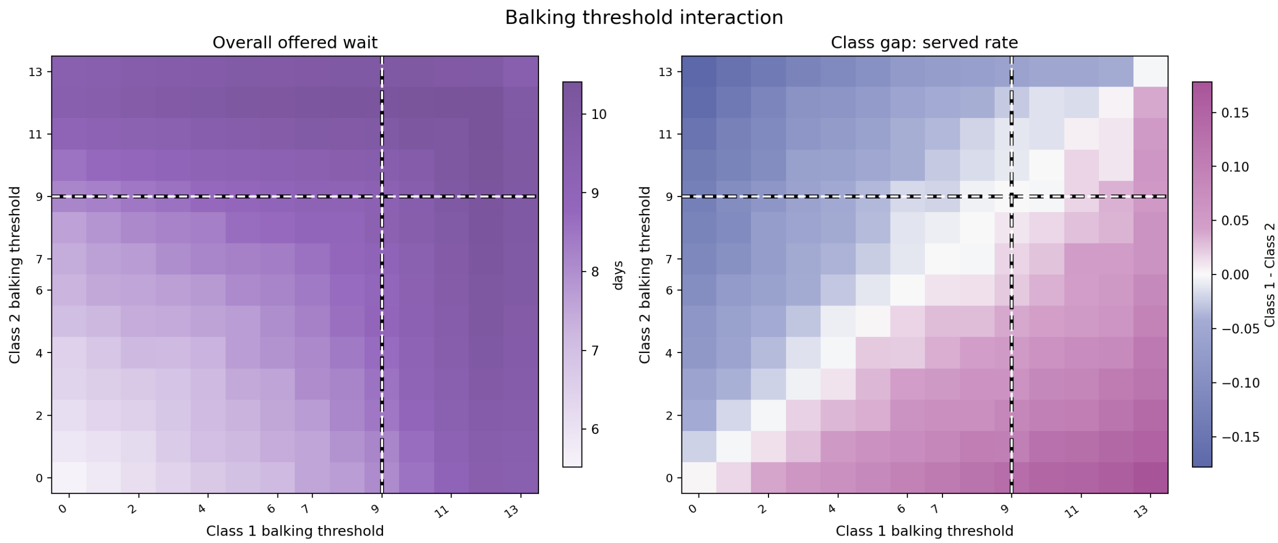

Figure 15: Balking threshold heatmap. Higher threshold allows longer waits before high-balking applies; the more tolerant class captures more served slots under FCFS.

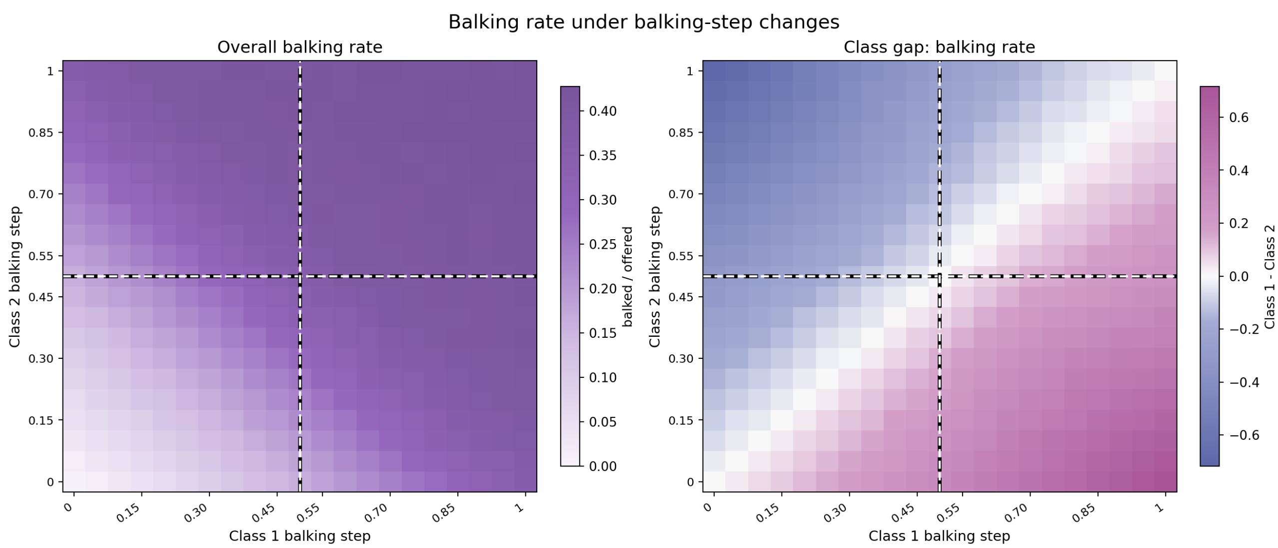

Figure 16: Balking rate under balking-step changes. The right panel shows the Class 1 minus Class 2 balking-rate gap.

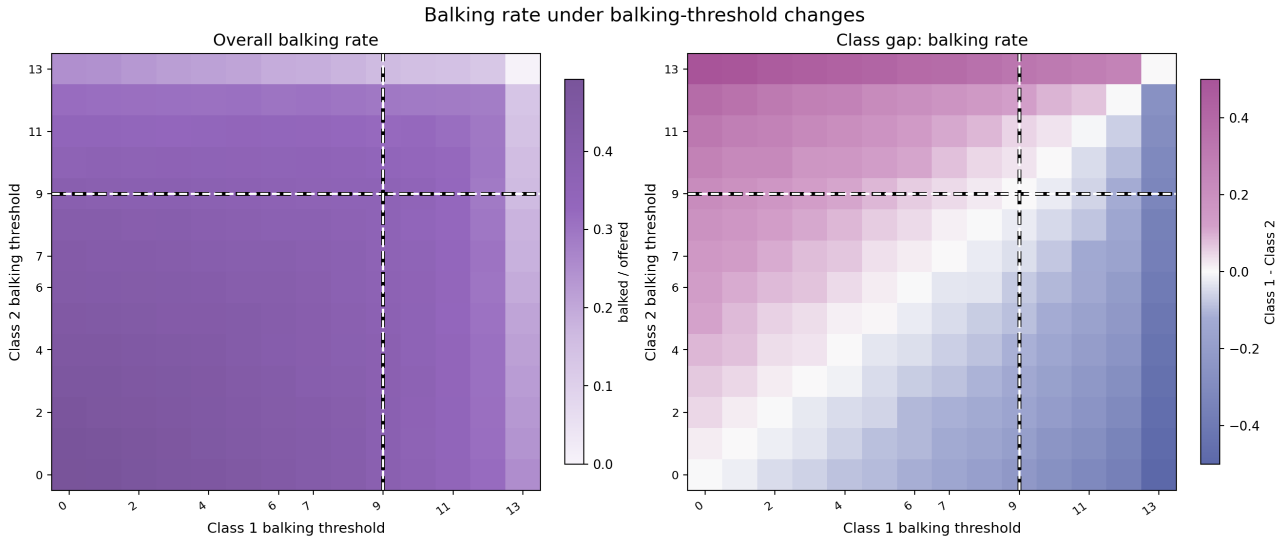

Figure 17: Balking rate under balking-threshold changes. The more patient class has the lower balking rate and the higher served rate.

A.3 Demand and Cancellation

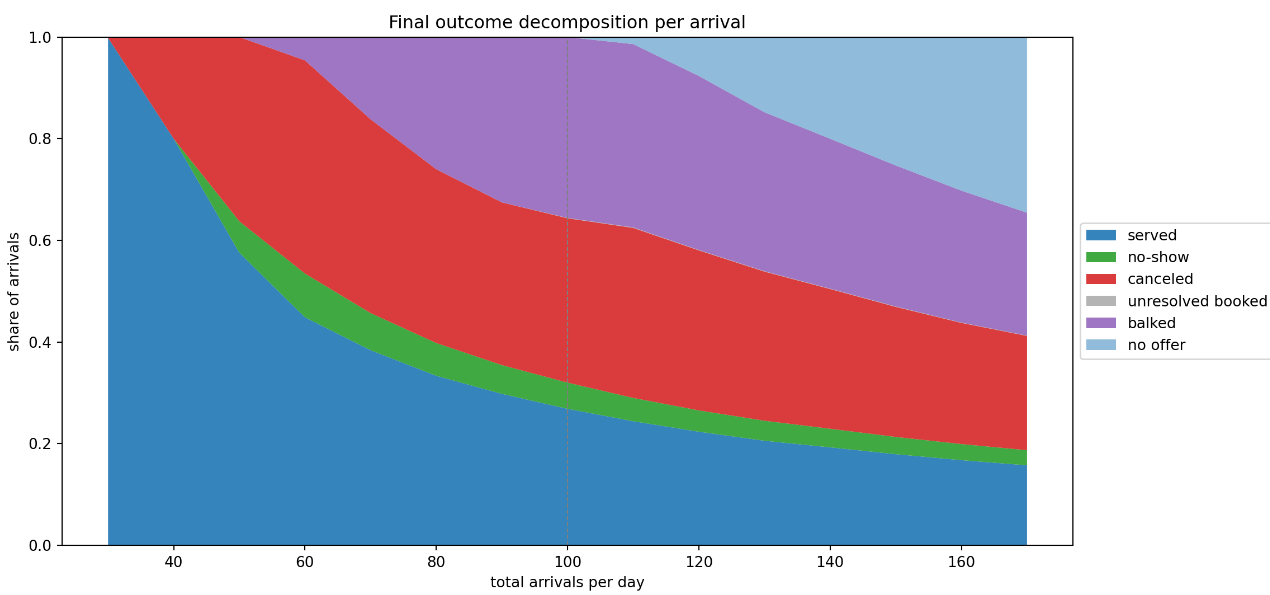

Figure 18: Outcome decomposition as total demand rises. At high demand, many arrivals receive no offer because the FCFS horizon fills; at moderate demand, balking and after-booking losses dominate.

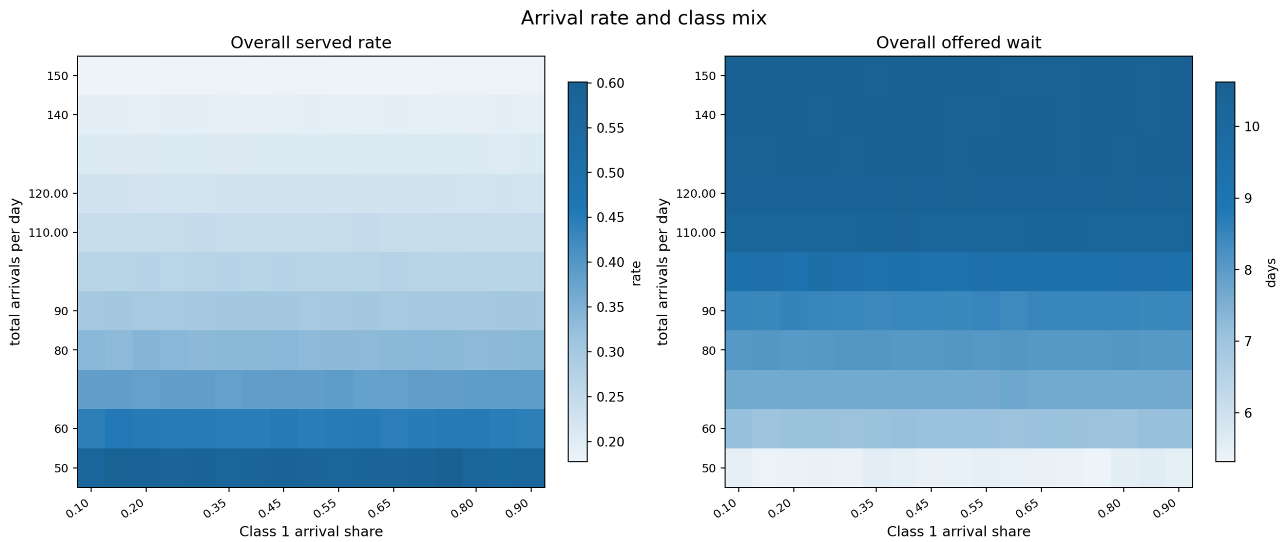

Figure 19: Arrival rate and class-mix heatmap. Left: overall served rate. Right: mean offered wait. Higher total demand worsens both.

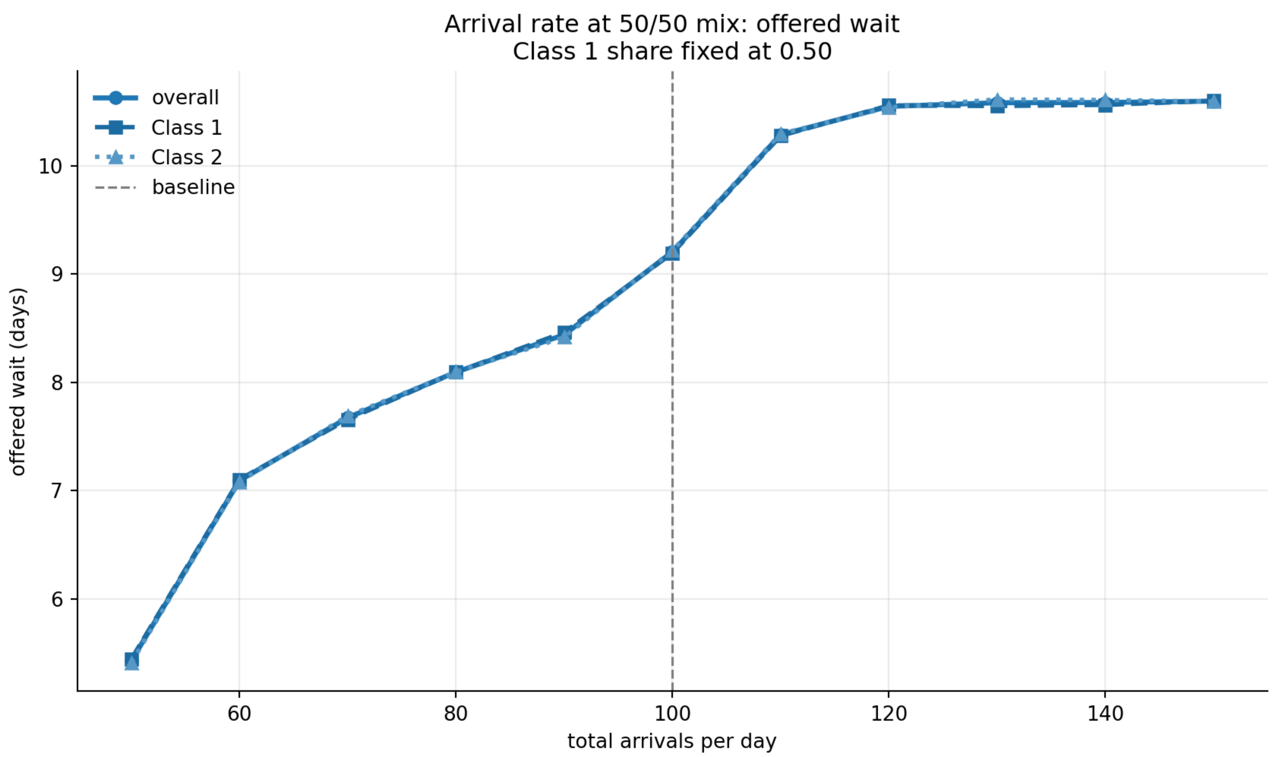

Figure 20: Mean offered wait as total arrivals change at a 50/50 class mix. Offered wait rises monotonically with demand.

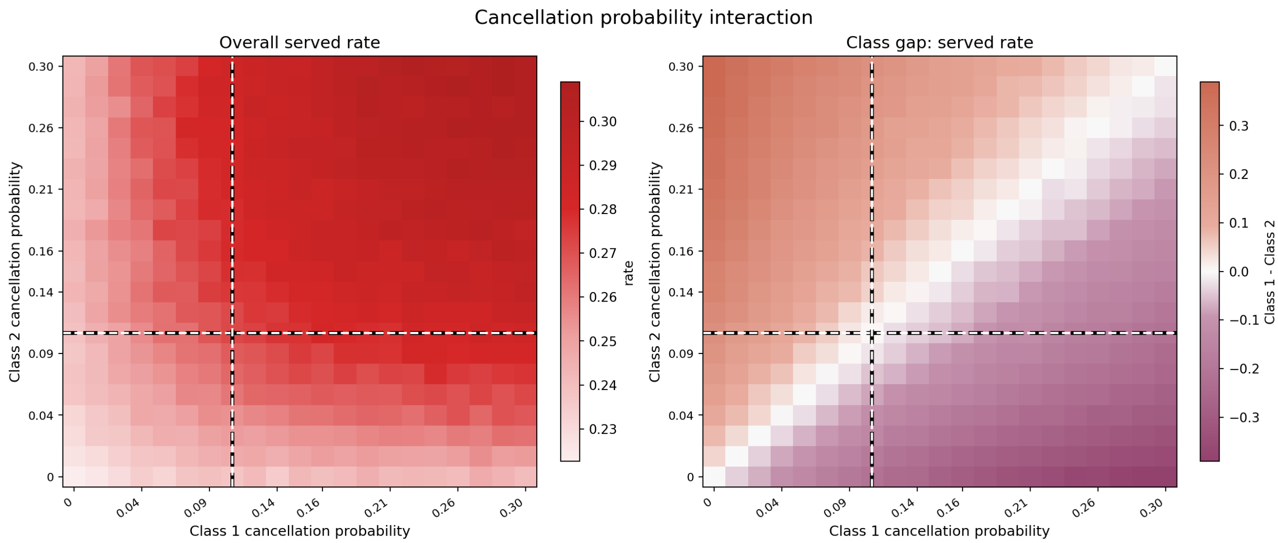

Figure 21: Cancellation heatmap. Left: overall served rate. Right: Class 1 served-rate gap. Higher cancellation for one class reduces that class's access.

A.4 Common Threshold Grids

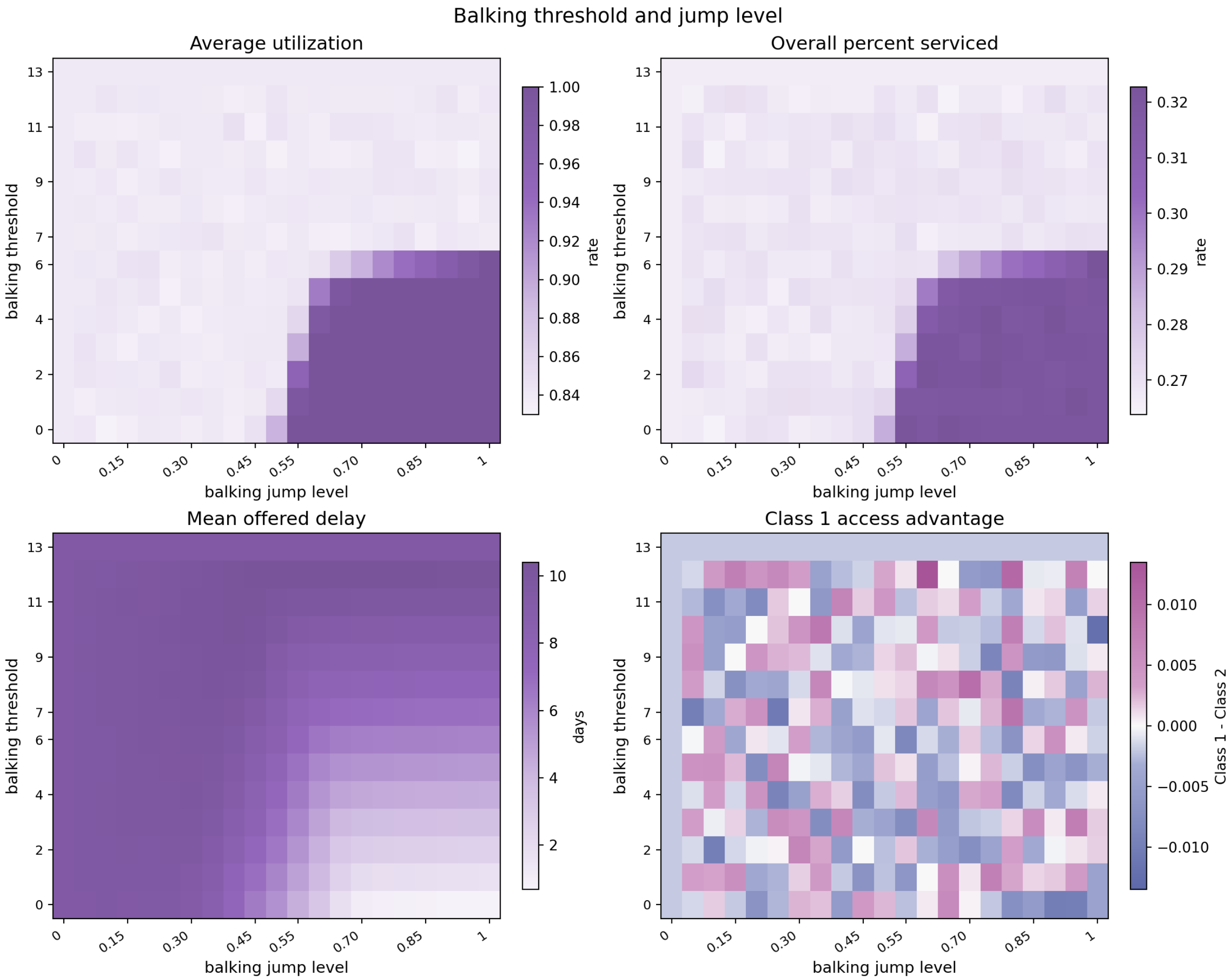

Figure 22: Shared balking threshold and jump level for both classes. These are symmetric aggregate tests, not class advantage tests.

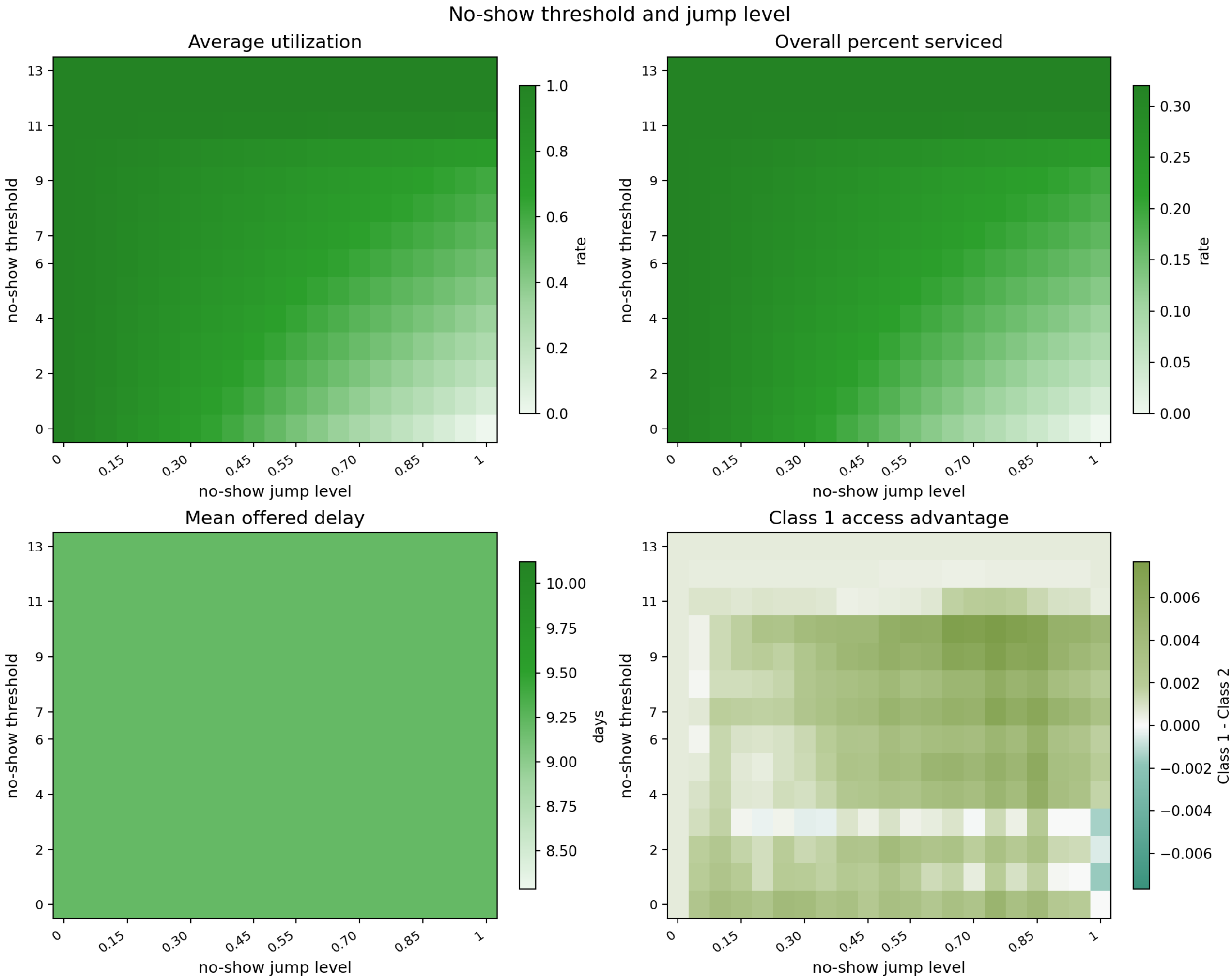

Figure 23: Shared no-show threshold and jump level for both classes.

A.5 Full Diagnostic Plots

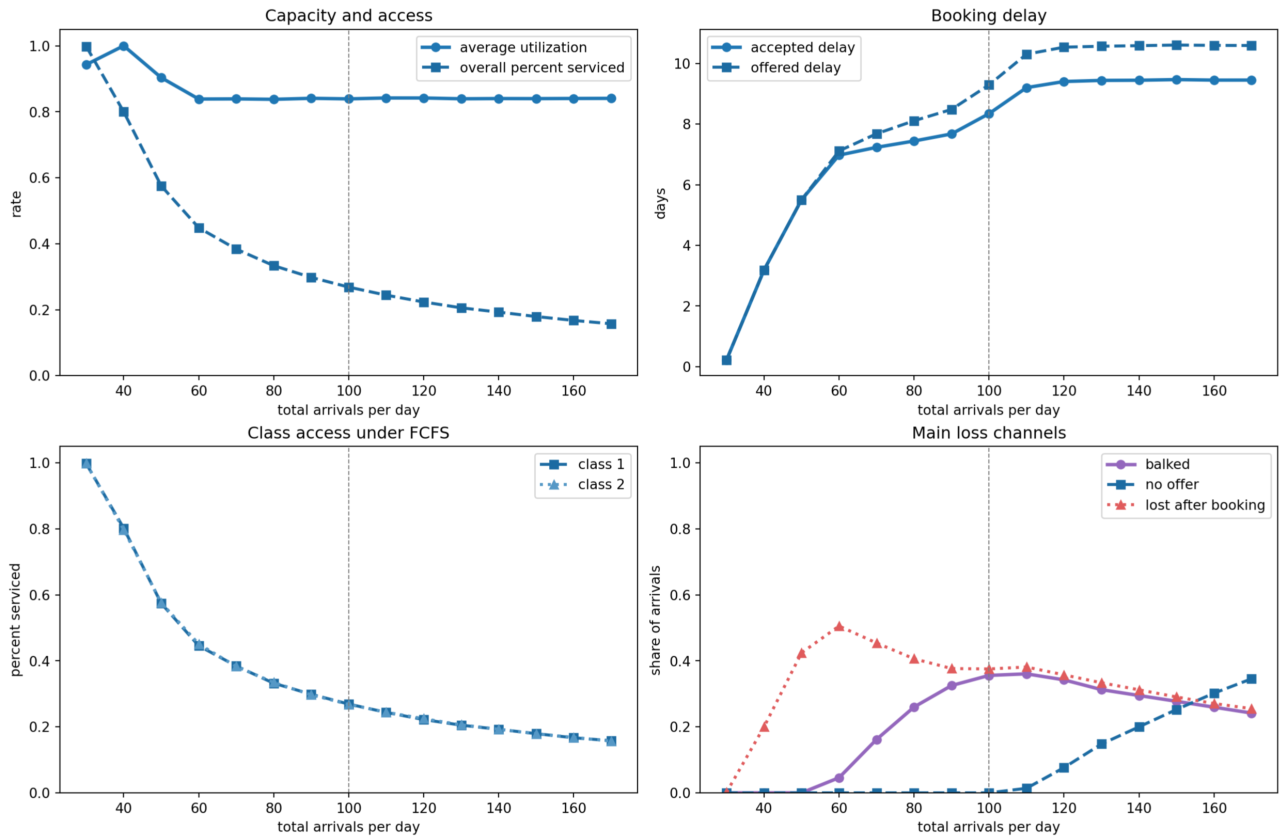

Figure 24: Full FCFS stress diagnostic plot across the wider arrival range.

A.6 Balking Deep Dive: Additional Plots

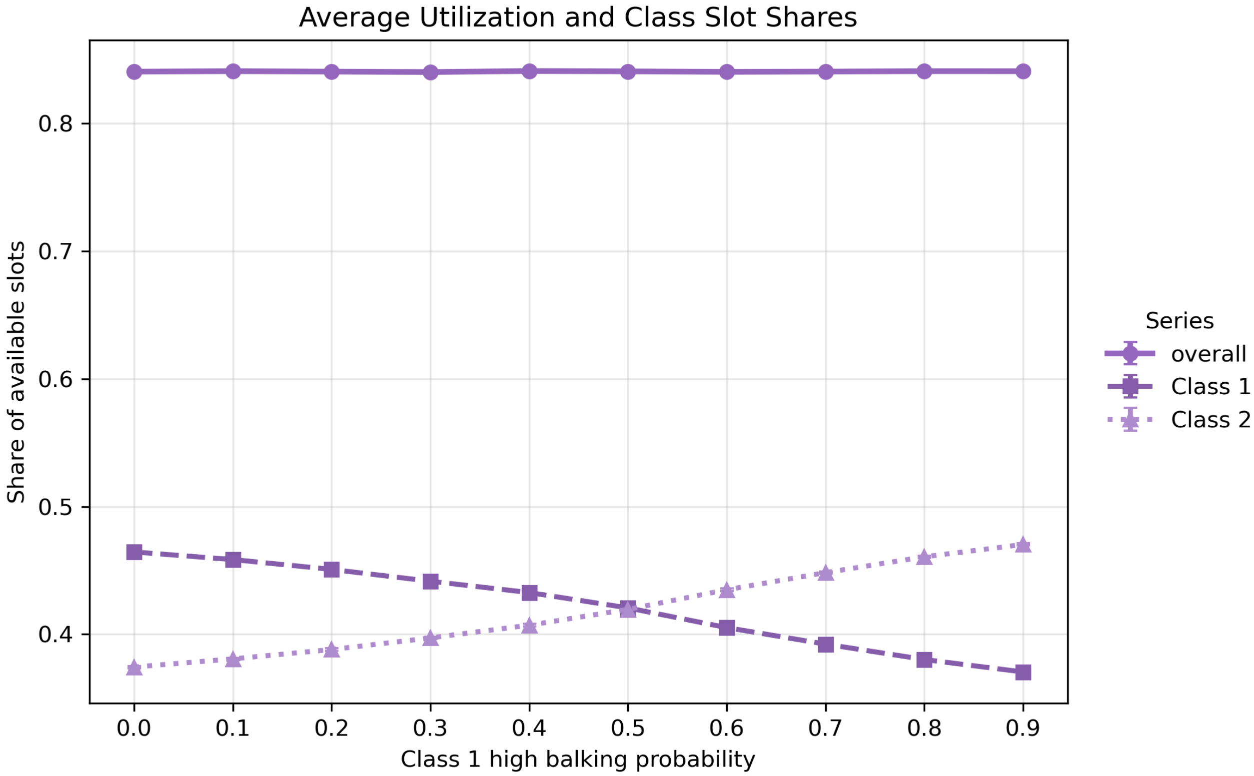

Figure 25: Class 1 high-balking-probability sweep: aggregate utilization remains nearly flat, indicating that Class 2 demand fills the slots vacated by balking Class 1 patients.

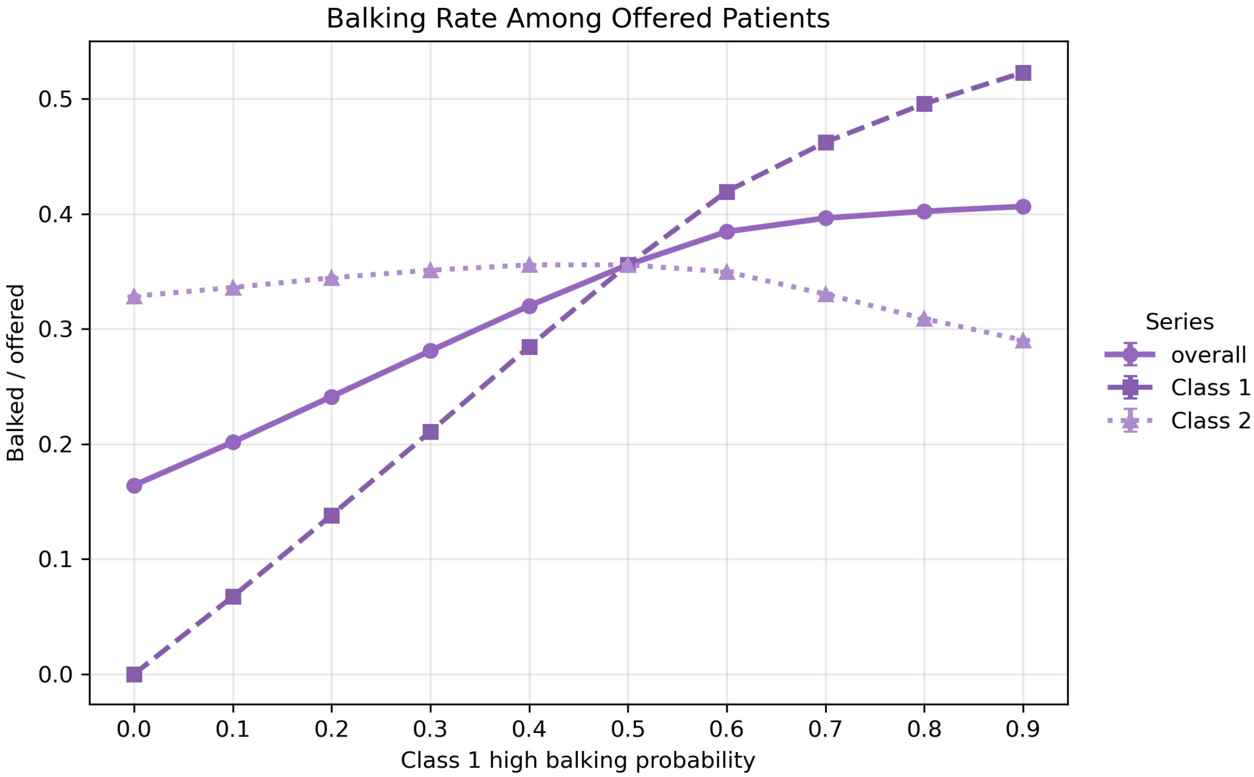

Figure 26: Class 1 high-balking-probability sweep: balking-rate confirmation that the mechanism is behavioral rejection, not capacity change.

Figure 27: Class 1 balking-threshold sweep: aggregate utilization is largely flat, confirming that threshold changes redistribute service without large aggregate effects.

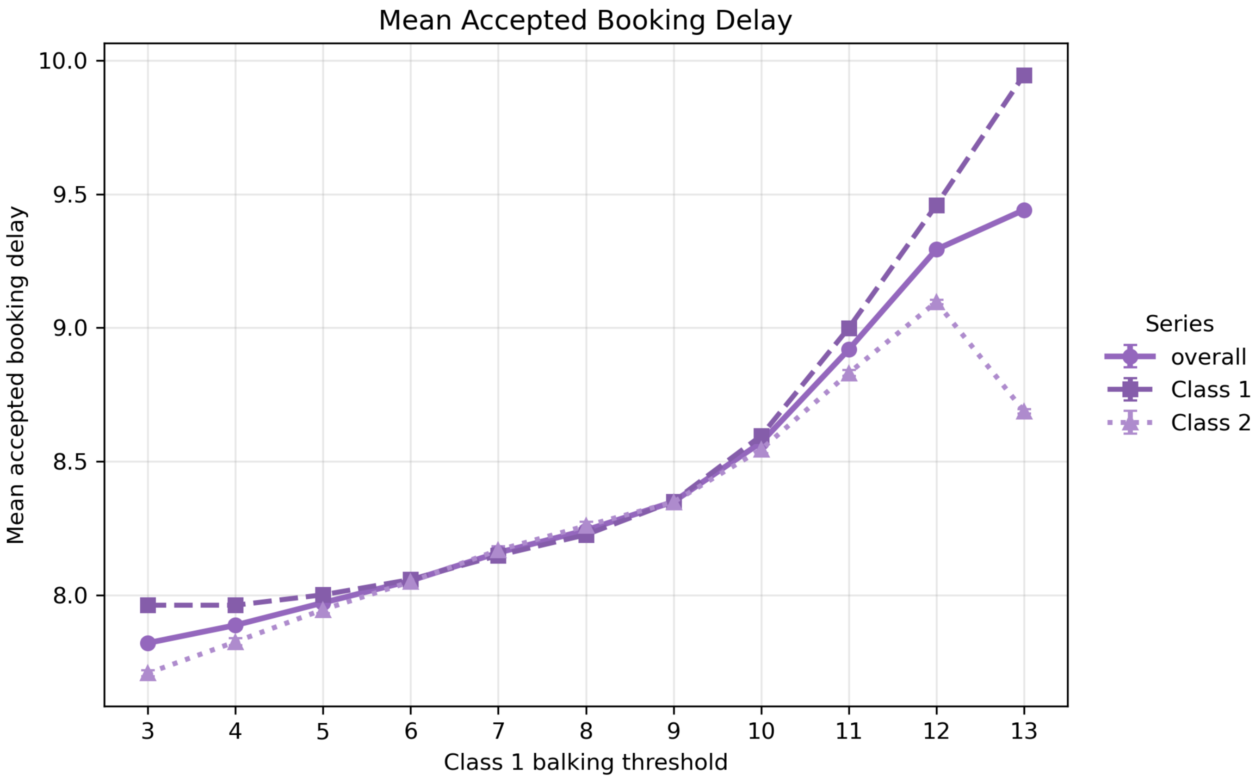

Figure 28: Class 1 balking-threshold sweep: Class 1 accepted delay rises with the threshold because patients tolerate longer waits before the high- balking probability applies.

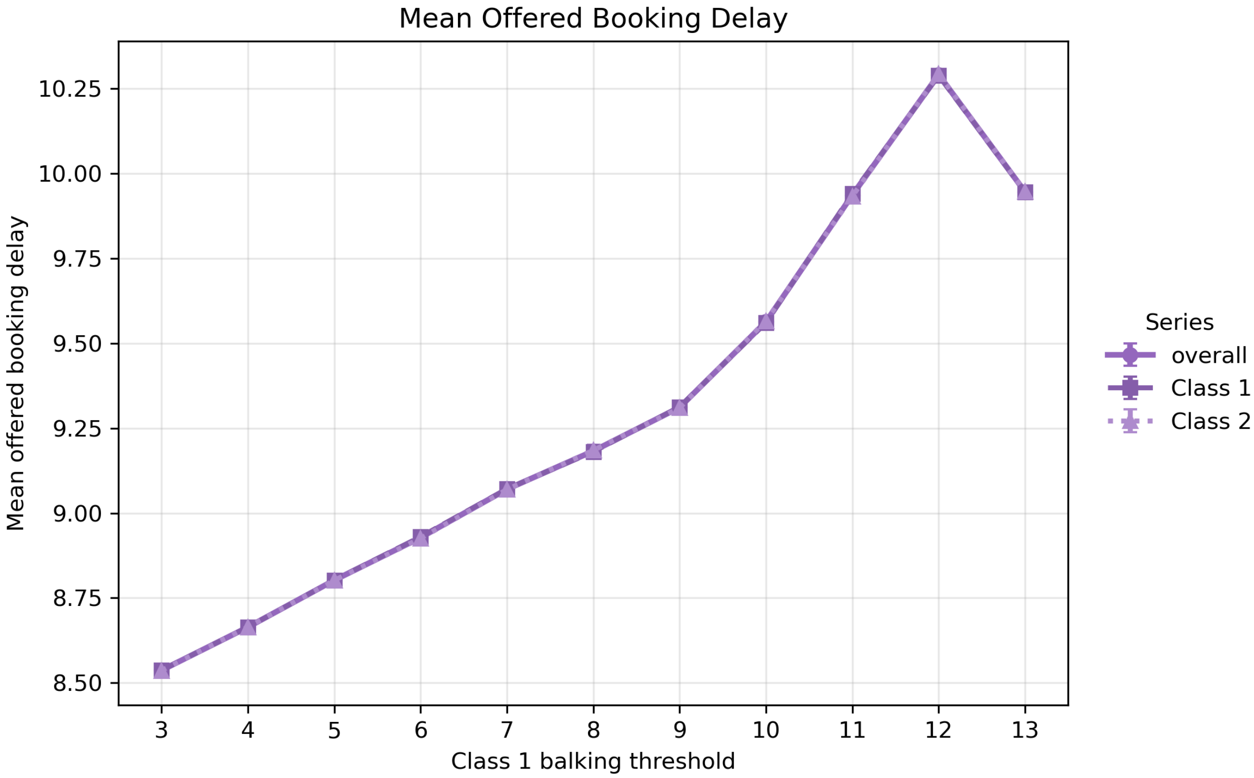

Figure 29: Class 1 balking-threshold sweep: mean offered delay by class.

B Appendix: Exact Metric Definitions

Let i be patient class, D be measured service days, S be slots per day, A_i be arrivals, B_i be accepted bookings, K_i be balked patients, C_i be cancellations, H_i be no-shows, V_i be completed visits, and $O_i = B_i + K_i$ be offered patients. Let $\bar{\tau}_i^{\text{offered}}$ be class i mean offered booking delay.

$$
\begin{aligned}
\texttt{percent\_serviced}_i &= \frac{V_i}{A_i}, \\
\texttt{overall\_percent\_serviced} &= \frac{\sum_i V_i}{\sum_i A_i}, \\
\texttt{mean\_offered\_booking\_delay} &= \frac{\sum_i \text{total offered delay}_i}{\sum_i O_i}, \\
\texttt{mean\_accepted\_booking\_delay} &= \frac{\sum_i \text{total accepted delay}_i}{\sum_i B_i}, \\
\texttt{average\_utilization} &= \frac{1}{D} \sum_d \frac{\text{served slots}_d}{S}, \\
\texttt{balking\_rate} &= \frac{\sum_i K_i}{\sum_i O_i}, \\
\Delta_{\text{access}} &= \texttt{percent\_serviced}_1 - \texttt{percent\_serviced}_2, \\
\Delta_{\text{delay}} &= \bar{\tau}_2^{\text{offered}} - \bar{\tau}_1^{\text{offered}}, \\
\Delta_{\text{balking}} &= \frac{K_1}{O_1} - \frac{K_2}{O_2}.
\end{aligned}
$$

Positive Δ_{access} means Class 1 has the higher served rate. Positive Δ_{delay} means Class 1 has the lower mean offered wait (Class 2 waits longer). Positive Δ_{balking} means Class 1 rejects offers more often.

Measurement semantics note. Class-level patient metrics are tracked for patients who arrive during the measurement window. If `cooldown_days` $<$ `horizon_days` -1, some late measurement-window bookings may remain unresolved when the simulation ends; these patients are currently not counted as served, canceled, or no-show. `total_value` is service-day based and is not cohort-aligned with the tracked class metrics.